Games Design Document

Project Chainy

Chainy: A 3D World Educational Game for learning emerging Technology (Blockchain & AI)



Figure 1. Exploration and guide interaction. Generated concept based on the approved Chainy references; not an implemented screenshot.

An offline educational exploration game

An offline Android educational game introducing foundational Web3, Blockchain and AI to learners aged 12-17, especially learners in low-income and remote communities with limited technology access.

Design edition: 22 October 2026. Based on the adopted October 12 technical design. Illustrations explain the intended experience; they are not evidence of a completed build. Exact mini-game rules and level count remain to be agreed.

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Reading this edition

This edition follows the sample GDD: numbered sections, a narrative brief, visual gameplay explanations, character descriptions and a project plan. The source sample supplies organization and layout, not RPG features or implementation claims.

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Current decision boundary

The October 12 replacement design governs this edition. Earlier nine-lesson maps, record-ordering simulations, three-copy puzzles, trained classifiers and grid agents are not required game systems.

1 Game objectives / brief

1.1 Game brief

An offline Android educational game introducing foundational Web3, Blockchain and AI to learners aged 12-17, especially learners in low-income and remote communities with limited technology access.

Single-player Unity/C# gameplay combines exploration, short learning interactions and three mini-games. Content, profiles and progress are stored locally. Internet access is not required during play.

1.2 Narrative brief

The learner joins a friendly technology learning community. A blue-haired explorer and screen-faced robots make unfamiliar ideas approachable through situations involving information, shared records and computer predictions. The precise story and sequence remain a design proposal.

1.3 Game objectives

Help learners recognize safe choices about digital information, describe what a shared record represents, and explain that AI systems can produce incorrect answers. Each activity must connect its puzzle to an explicit concept and explanation.

Engagement comes from exploring, making choices, solving a short challenge and understanding the result. Completing an activity or enjoying it does not by itself demonstrate learning improvement.

Key features

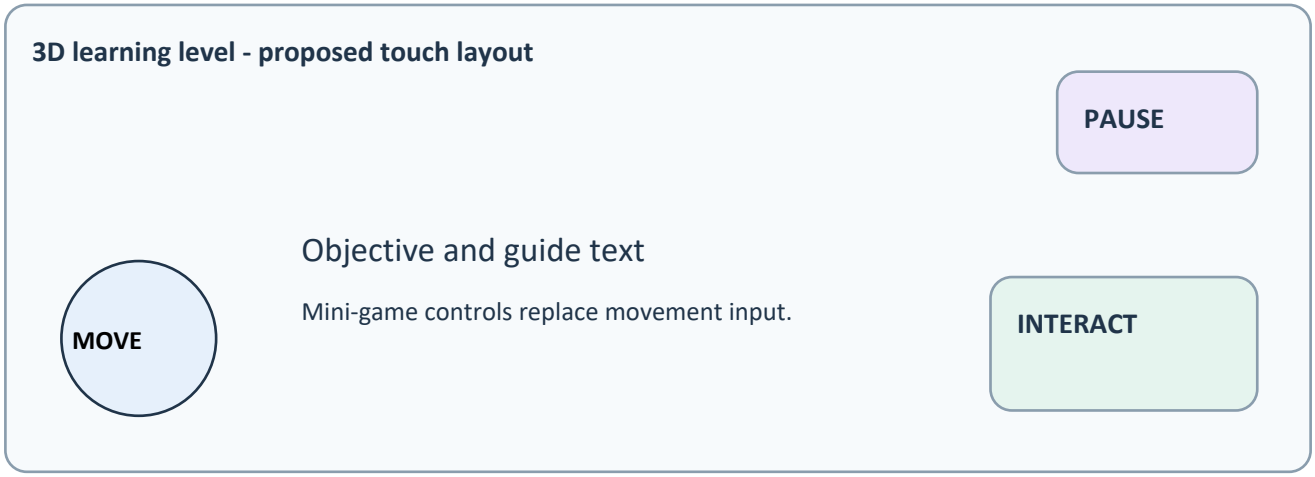
Offline play; compact 3D learning areas; Quiz, Wordle and BlockMatch; readable explanations; separate profiles on shared devices; locally saved progress.

2 Gameplay

2.1 Overview

You enter a small island learning area with the blue-haired explorer. A friendly robot asks you to inspect a community record and explains why people keep copies. You start a Quiz, choose an answer and read why it is correct or needs another look. In another activity, you guess a technology word and connect it to its definition. A BlockMatch board gives you a matching objective and explains the learning connection after the result. Only one mini-game is active at a time. After saving, you return to the level or leave; the same device can load your saved progress later.

2.2 Layout and controls



Proposed control arrangement for touch testing, not a verified screen. Exact movement and camera tuning remain team decisions.

Begin with a short unscored control introduction. Movement and interaction operate in the level. When a mini-game opens, only its inputs are active. Pause offers a clear way to resume or leave.

The result screen explains the outcome and available next action. Failed saves show retry rather than claiming success. A learner code selects progress on this device; it is not an online login.

3 Game world

3.1 Learning environment brief



Figure 2. Proposed island atmosphere and guide station. Scene scale, camera and route are illustrative.

Use the approved colorful, rounded island scenery and existing player/world assets. A representative compact area should establish movement, dialogue and entry into one activity before scene quantity expands.

3.1.1 Learning locations

Proposed area	Purpose
Arrival point	Select the learner profile and practice basic controls.
Guide interaction	Read a short explanation and inspect a concrete example.
Activity point	Open one approved mini-game; return after its result.
Progress view	See saved work and choose the next available activity.

Content themes

Web3: information, control, privacy and safe choices. Blockchain: records, links and shared history. AI: examples, predictions, mistakes and responsible checking. These are topic recommendations for the content/rules sheet, not a mandatory number of worlds or levels.

4 Graphics / art style / sound design

4.1 Storyboard and GUI development

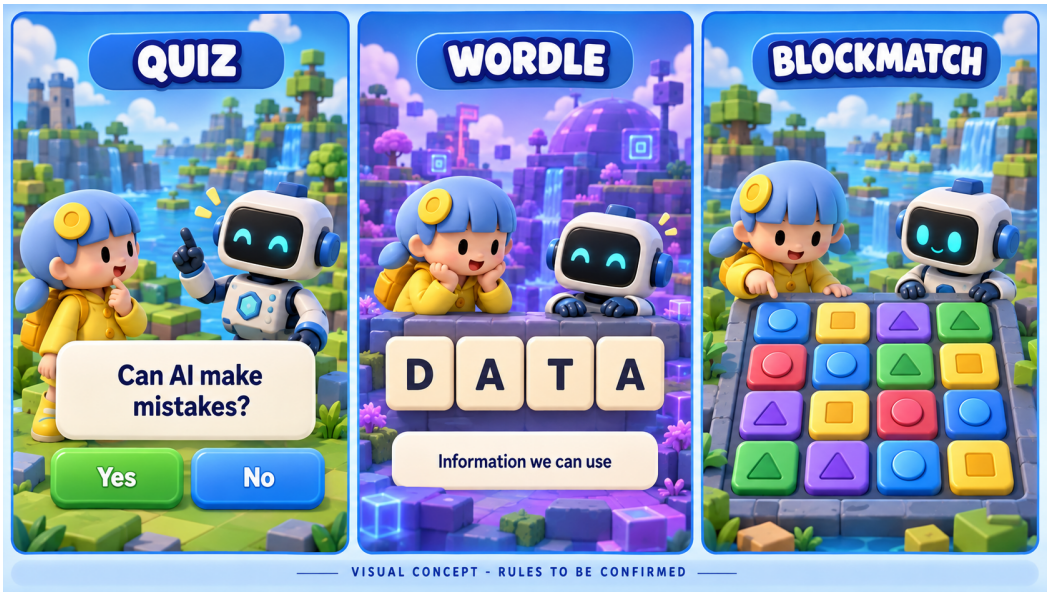


Figure 3. Separate activity concepts. DATA is an example, not a fixed word length. Tile count, colors and board rules are illustrative; only one activity runs at a time.

The learner enters a level, reads the guide, opens one activity, acts, reads feedback, and saves before returning. UI screens include profile, menu, level, instructions, activity, result, pause and save-error states.

4.1.1 Art direction

Retain the blue-haired explorer in yellow, friendly robots, rounded toy-like 3D forms and readable island scenery. Use color to support meaning, with text or symbols alongside it. Earlier concept images are style references; they do not approve old gameplay.

4.2 Example narrative development

Proposed guide exchange: "Our club keeps a record of borrowed books. What does this entry tell us?" After the learner chooses: "It records a claim about a loan. We still need reliable information about what happened." The example introduces a Quiz question without implying records prove real-world truth.

4.3 Sound

Short selection and result sounds may support feedback. Optional gentle music should remain unobtrusive. All essential instructions and results work with sound muted; narration is not a required system.

5 Mechanics

5.1 Learning interactions

Move through the level, interact with a guide, inspect the concept and open a mini-game. A displayed objective tells the learner what to do. An explanation connects the result to the subject matter.

5.2 Quiz, Wordle and BlockMatch

Mode	Established behavior	Still to agree
Quiz	Questions, answer evaluation and scoring.	Question set, score/pass rule and retries.
Wordle	Guess validation and letter feedback.	Dictionary, word length, attempts, repeated letters and end rules.
BlockMatch	Board moves, matches, cascades and objectives.	Match meaning, board limits, move/time limit, no-move and end rules.

5.3 Progression and outcomes

A mini-game reports its outcome using the agreed rules. Success may enable the next learning step; a mistake or exhausted attempt allowance offers the agreed retry or exit path. Exact unlocks and score thresholds remain open. No health, combat or currency system is required.

5.4 Saving and input

Only one mini-game accepts input. Loading, saving and resolving a board prevent duplicate submissions. SQLite stores local records. Show completion as saved only after a successful write. Preserve the last good record if saving fails.

Educational content about AI does not require a working AI model. Wordle needs a curated educational word list and definitions. BlockMatch needs a stated educational connection, not only visual matching.

6 Characters



Existing concept interpretation. Original approved storyboards remain the authority for proportions and costume.

6.1 Character mechanics

The explorer moves and interacts with the learning environment. The learner controls selections, guesses and board moves during activities. Friendly robots provide explanations; character names and individual roles remain proposed.

6.2 Character development and animation

Use a small set of readable movements: idle, move, greet, indicate an object and acknowledge a result. Animation should not obscure text or delay a correction. Combat and weapon animations are not applicable.

6.3 Player properties

Property	Visible feedback
Active learner	Profile label changes; previous learner data is cleared from the screen.
Position / interaction	Explorer moves; a nearby interaction becomes available.
Active mini-game	Only its controls and current objective receive gameplay input.
Answer, guess or board	Selection, letter feedback or match result reflects the action.
Result and saved progress	Outcome is explained; saved confirmation appears only after a successful write.

7 Project plan

7.1 Team ownership

Sherwin owns coordination and game-design decisions. Christian owns game code, data, builds and integration. Luis owns art assets. Sean authors project documents. Miguel owns game UI implementation. Cross-role work passes through a separate handoff card; cards have one named owner.

7.2 Delivery schedule

Date / window	Planned evidence
October 18 - prior target	First Quiz, save and reload loop; acceptance status must be checked.
October 25	Three modes and agreed learning content ready for integration.
October 26-30	Integrated candidate and beta feature freeze.
October 31-November 2	Android checks and beta readiness.
November 3-6, 2026	Fixed beta at Tambunan National High School, Marinduque.
November 7-16	Actual results, fixes and manuscript completion.
November 17-20	Rehearsal and final preparation.
November 21, 2026	Fixed pre-final defense.

7.3 Progress reporting

The team board records current work and named history; GitHub receives queued issue copies. Dates are planning targets, not evidence of completed work. Actual device readiness and school arrangements still need records. The supplied 32-point rubric does not define how to calculate the school's 60-percent completion requirement.

8 References

The user-supplied Project Vanishing GDD provides the numbered presentation format. Original Chainy storyboards supply visual direction. Christian Jude Villaber's October 12 technical design supplies current gameplay and system facts. The project baseline and task plan record the adopted decisions and fixed dates.

9 Bibliography and supporting records

Project Vanishing. Games Design Document. User-supplied 27-page example; author/date not independently verified.

Game Design Document Template. User-supplied six-page template; used for the companion edition.

Christian Jude Villaber. Chainy technical design. Adopted October 12, 2026. Canonical source: docs/archive/source-material/technical-design-christian-2026-10-12.md.

Chainy project baseline; current game design; current task plan. Repository documents consulted October 22, 2026.

Technical target

Android 10+/11+, 4 GB physical RAM, 64 GB total device storage, ARM64, OpenGL ES 3.2 or Vulkan 1.1, entry-level-to-midrange GPU.

Unity/C#, mobile touch controls, local SQLite. Android 11+ is the provisional test starting point; Android 10 is a compatibility candidate. Support is unverified, and total phone storage is not the app size.

Rule decisions before content freeze

Freeze exact Quiz scores and completion, Wordle words and guess feedback, BlockMatch matches and endings, and the meaning of saved score/attempt/progress. These editions do not silently decide those open rules.